

PROJECT 5: CREATE IAM USERS, GROUPS, AND POLICIES REPORT

INTRODUCTION

AWS Identity and Access Management (IAM) is a core security service provided by Amazon Web Services (AWS) that enables organizations to securely manage access to AWS resources. IAM allows administrators to create users, organize them into groups, and assign permissions through policies, ensuring that only authorized individuals can access specific services and resources.

This project focuses on implementing IAM by creating an IAM user, creating an IAM group, attaching a policy to the group, assigning the user to the group, and verifying access permissions. The project demonstrates how AWS IAM can be used to establish secure, scalable, and policy-driven access control within a cloud environment.

OBJECTIVES

The primary objectives of this project are:

- ❖ Create an IAM user
- ❖ Create an IAM group
- ❖ Attach a policy to the IAM group
- ❖ Add the IAM user to the group
- ❖ Log in using IAM user credentials
- ❖ Verify assigned permissions and access restrictions
- ❖ Understand the fundamentals of AWS Identity and Access Management (IAM)

OVERVIEW OF AWS IDENTITY AND ACCESS MANAGEMENT (IAM)

AWS Identity and Access Management (IAM) is a web service that helps administrators securely control access to AWS resources. IAM provides mechanisms for authentication and authorization, allowing organizations to manage who can access AWS services and what actions they can perform.

IAM plays a critical role in cloud security by enforcing access controls and implementing the principle of least privilege.

Key Concepts

IAM User

An IAM User is an individual identity created within an AWS account. Each user can have unique credentials and permissions.

IAM Group

An IAM Group is a collection of IAM users. Permissions assigned to a group are automatically inherited by all users within that group.

IAM Policy

An IAM Policy is a JSON-based document that defines permissions and specifies which actions are allowed or denied on AWS resources.

Authentication

Authentication is the process of verifying the identity of a user attempting to access AWS resources.

Authorization

Authorization determines the actions and resources a user is permitted to access after successful authentication.

Principle of Least Privilege

The Principle of Least Privilege ensures that users receive only the permissions necessary to perform their required tasks, reducing security risks.

METHODOLOGY

Step 1: Creating the IAM User

An IAM user named **cloudintern-user** was created using the AWS Management Console. Console access was enabled, and password authentication was configured to allow secure login through the AWS Management Console.

This user represents an individual identity that can access AWS resources based on assigned permissions.

Step 2: Creating the IAM Group

An IAM group named **S3-ReadOnly-Group** was created to simplify permission management.

Using groups allows administrators to assign permissions once and apply them to multiple users, reducing administrative overhead and improving consistency.

Step 3: Attaching the IAM Policy

The AWS-managed policy **AmazonS3ReadOnlyAccess** was attached to the **S3-ReadOnly-Group**.

This policy grants users permission to:

- ❖ View Amazon S3 buckets
- ❖ List bucket contents
- ❖ Read objects stored in S3
- ❖ Access S3 resources without modification privileges

The policy does not allow users to create, modify, or delete S3 resources.

Step 4: Adding the User to the Group

The IAM user **cloudintern-user** was added to the **S3-ReadOnly-Group**.

By becoming a member of the group, the user automatically inherited all permissions granted through the attached policy.

This demonstrates the effectiveness of group-based permission management in AWS.

Step 5: Logging in as the IAM User

The newly created IAM user successfully logged into the AWS Management Console using IAM credentials.

During the first login process:

- ❖ The user authenticated successfully
- ❖ Password reset procedures were completed
- ❖ Access to AWS services was established according to assigned permissions

Step 6: Verifying User Permissions

Permission verification was performed to ensure that access controls were functioning correctly.

Amazon S3 Access

The IAM user successfully accessed Amazon S3 resources and was able to view available buckets and objects as permitted by the attached policy.

Amazon EC2 Access

Attempts to access Amazon EC2 services resulted in authorization errors because no EC2 permissions were assigned to the user.

This confirmed that IAM policies were correctly enforcing access restrictions.

RESULTS

The following outcomes were successfully achieved:

- ❖ Created an IAM user named **cloudintern-user**
- ❖ Created an IAM group named **S3-ReadOnly-Group**
- ❖ Attached the **AmazonS3ReadOnlyAccess** policy to the group
- ❖ Added the IAM user to the group
- ❖ Successfully logged in using IAM credentials
- ❖ Verified policy-based access control
- ❖ Confirmed read-only access to Amazon S3 resources
- ❖ Confirmed restricted access to unauthorized AWS services such as Amazon EC2

KEY LEARNINGS

Throughout this project, several important IAM concepts were learned:

- ❖ IAM provides secure identity and access management for AWS resources
- ❖ Groups simplify permission assignment and administration
- ❖ Policies define what actions users can perform
- ❖ Users inherit permissions from assigned groups
- ❖ Access restrictions can be enforced using IAM policies
- ❖ The Principle of Least Privilege enhances cloud security
- ❖ IAM is a foundational component of AWS security and governance

ADVANTAGES OF USING AWS IAM

AWS IAM offers several benefits for organizations and cloud administrators:

- ❖ Centralized user and permission management
- ❖ Enhanced security through controlled access
- ❖ Simplified administration using groups and policies
- ❖ Support for the Principle of Least Privilege and Improved compliance and governance
- ❖ Fine-grained access control across AWS services and Seamless integration with all AWS resources and services

LIMITATIONS OF AWS IAM

Despite its advantages, IAM also presents certain challenges:

- ❖ Misconfigured policies may grant excessive permissions
- ❖ Complex permission structures can become difficult to manage
- ❖ Large-scale environments require careful planning
- ❖ Human errors can impact security and access control
- ❖ Regular auditing and monitoring are necessary
- ❖ Understanding policy inheritance may require additional learning

CONCLUSION

This project provided practical hands-on experience with AWS Identity and Access Management (IAM). The implementation involved creating IAM users, creating IAM groups, attaching policies, assigning permissions through group membership, and validating access restrictions.

The successful completion of this project demonstrated how IAM enables secure and efficient access management within AWS environments. It also reinforced the importance of implementing the Principle of Least Privilege, managing permissions through groups, and using policies to enforce security controls.

Overall, the project strengthened understanding of AWS security fundamentals and highlighted the critical role of IAM in protecting cloud resources and maintaining organizational security.